

①

* Derivatives :-

→ Derivatives of constant is 0.

For ex :-

$$\frac{d}{dx} [c] = 0, \quad \frac{d}{dx} [5] = 0, \quad \frac{d}{dx} [-7] = 0$$

$$\rightarrow \frac{d}{dx} [f(x)] = f'(x)$$

o Power Rule :-

$$\rightarrow \frac{d}{dx} (x^n) = nx^{n-1}$$

For ex :-

$$\frac{d}{dx} (x^2) = 2x, \quad \frac{d}{dx} (x^4) = 4x^3$$

o Constant Multiple Rule :-

$$\rightarrow \frac{d}{dx} [c \cdot f(x)] = c \cdot \frac{d}{dx} [f(x)]$$

For ex :-

$$\begin{aligned} 1) \quad \frac{d}{dx} [4x^7] &= 4 \cdot \frac{d}{dx} [x^7] \\ &= 4 (7x^6) \\ &= 28x^6 \end{aligned}$$

(2)

• Finding derivatives using limit Process.

$$\frac{d}{dx} f(x^2) = 2x \rightarrow \text{How?}$$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)(x+h) - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{x^2} + xh + xh + h^2 - \cancel{x^2}}{h} \\ &= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{h}(2x+h)}{\cancel{h}} \\ &= \lim_{h \rightarrow 0} 2x + h \end{aligned}$$

$\therefore$ while h approach to 0 [$h \rightarrow 0$] then
 h become 0 [$h = 0$]

$$= \lim_{h \rightarrow 0} 2x$$

$$f'(x) = \lim_{h \rightarrow 0} 2x$$

$$f'(x) = 2x$$

$$f(x) = x^2$$

Then

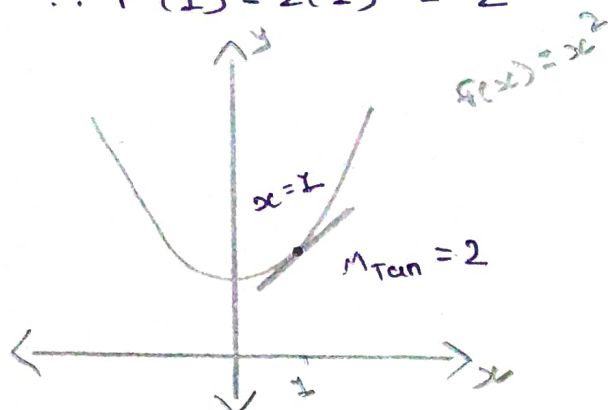
$$f(x+h) = (x+h)^2$$

(3)

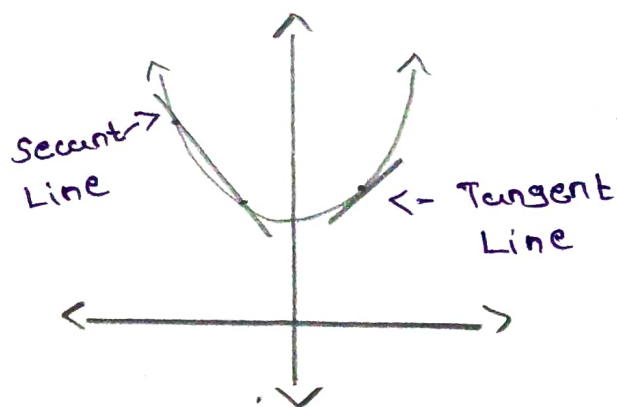
• Slope of The Tangent Line

$$f(x) = x^2 \quad f'(x) = 2x \quad \dots \text{ at } x = 1$$

$$\therefore f'(1) = 2(1) = 2$$



• Secant line & Tangent Line



1. Tangent line is a line That touches the curve at one Point.

2. Secant Line is a line That touches the curve at two Points.

$$m_{\text{sec}} = \frac{y_2 - y_1}{x_2 - x_1}$$

• Derivatives of Polynomial function

for Ex:-

$$f(x) = x^3 + 7x^2 - 8x + 6$$

$$f'(x) = 3x^2 + 14x - 8 \neq 0$$

$$f'(x) = 3x^2 + 14x - 8$$

Derivatives of Trigonometric functions

1. $\frac{d}{dx} [\sin x] = \cos x$
2. $\frac{d}{dx} [\cos x] = -\sin x$
3. $\frac{d}{dx} [\sec x] = \sec x \tan x$
4. $\frac{d}{dx} [\csc x] = -\csc x \cot x$
5. $\frac{d}{dx} [\tan x] = \sec^2 x$
6. $\frac{d}{dx} [\cot x] = -\csc^2 x$

Product Rule

$$* [f \cdot g]' = f' \cdot g + f \cdot g'$$

Ex:-

$$1. \frac{d}{dx} [x^2 \sin x] = 2x (\sin x) + x^2 (\cos x)$$

$$* [f \cdot g \cdot h] = f'gh + fg'h + fgh'$$

Ex:-

$$\begin{aligned} \frac{d}{dx} [x^3 (\tan x) (3x^2 - 9)] \\ = 3x^2 (\tan x) (3x^2 - 9) + x^3 (\sec^2) (3x^2 - 9) + x^3 (\tan x) (6x) \end{aligned}$$

$$* \frac{d}{dx} \left[\frac{f}{g} \right] = \frac{g f' - f g'}{g^2}$$